已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

Module Code	Module Name	Autumn	Spring
Compulsory Project Modules			
06-26581	Computer Science Project	20	20

- Mandatory for everyone (except MSc students, but read this as well)x
- Like a dissertation for other subjects
- Project meetings with your supervisor roughly every week (most of the time in groups, though that's up to each supervisor to decide)
- Later this term you will pick 5 potential supervisors and rank them in preference order (the supervisors will do the same), then Gale-Shapley (with a variation for differently sized sets) will be used to allocate you to a supervisor
- **Important!** Don't feel any pressure to pick a project now - it probably won't help you get ahead or anything. Instead, have some rough ideas of areas you might like to look into, and select a supervisor that will be able to help you in those areas.
    - Often supervisors will have a list of potential topics to research if you don't know what you want to do by the start of your project (also they're good because the supervisor will definitely know something about them and be invested in them)
    - Take a look at potential supervisors' research interests on their CS website

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

shortlist

- **MATHS STUDENTS: this is not mandatory for you (more maths advice @ end)**
  - You can choose to do EITHER this OR Research Skills in Mathematics
  - One or the other. Not both, not neither
    - Actually I think you can take both if you ask nicely but don't. Just don't. It's not worth it
  - CS project is a lot more involved and kinesthetic, maths projects usually more theoretical and more reading
  - Which one you pick changes how many credits you can take from each module so take that into consideration
  - Also maths project is only 20 credits and cs project is 40, so it's worth more if you do well, but fewer options for other modules

06-26587	Project Masters M60	20	40

- The mandatory module instead of Computer Science Project for MSci students
- Basically the same as Computer Science project
- Will be expected to be a bit bigger project due to having 20 more credits
- Presentation at the end of first term about your project, normally discussing the literature you've read and your plans of what you will do - pre-COVID this was a poster presentation, this year it was a video

Optional Modules (LH)

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

	massively delayed this year meaning students had to take their final exam without any feedback on how they had done in the module. Be cautious		
06-35308	Algorithms and Complexity	-	20
06-35326	Algorithms and Complexity (Extended)	-	20
	• Discusses a lot of complexity and calculating the time and space complexity of algorithms as well as proving correctness • If you enjoyed Logic and Computation, you may enjoy this • Good mix between abstract theory (P vs NP, NP and coNP, hierarchy theorems) and practical algorithm design (stable matching problem, sorting algorithms, lots of graph manipulation) • Gives you weekly formatives that you don't have to hand in, just check the answers for yourself, which is nice if you get anxious getting marks back • The class tests are basically worksheets - LaTeX is useful for this • Lectures have a bad habit of being either 'here's a super easy example, now you can do this for anything' or 'here's some simple stuff, now do a lot harder stuff' • Expects a lot of background knowledge on the topic ○ If you liked the stuff on Complexity and Turing Machines in Maths and Logic for CS, you may like this module • Comes across very undirected, and goes over quite a lot of material more than once for no reason		
06-30213	Computer Vision and Imaging	-	20

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

is negative there has been no chance for the module to improve yet.
- The content relies heavily on maths. The lecturers give basic outline to the maths content, but sometimes you have to spend time researching on your own to fully understand it. If you are still having problems, the lecturers and TAs organise office hours 5 times a week, where they give a very good explanation to module related questions.
- Even though it is not mentioned in the title of the module, the content is heavily linked to AI, and I think it gives a good appreciation of what computers can do nowadays.
- If it's anything like the old Computer Vision 2nd year module:
    - Weekly formative assignment using MATLAB
    - Hamid was a pretty good lecturer and gives good feedback if you ask for it - He's not teaching it anymore though?
    - Involves some maths in the form of Matrix Multiplication
    - Quite a bit of theory?
    - Also one week on Human vision that is assessed content (knowing how visual signals travel from your eye to your brain, rods & cones, etc.)
- Had poor feedback on student rep survey
    - Students not receiving clear and justified feedback on work
    - Lectures rushed
    - Assignment didn't fit the assessment structure leading to it feeling rushed
    - Formative assessments much smaller and easier

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

06-35376	Evolutionary Computation (Extended)	-	20

• no
  ○ just don't
  ○ Big yikes
• Any useful information you can provide?
  ○ It's one of the modules where it seems really interesting to learn about but really horrible to then be assessed on
  ○ Another AI module, there's better ones to take
  ○ If it's anything like its precursor module there was some interesting stuff on constraints			

06-30214	Human-Computer Interaction	20	-
06-30251	Human-Computer Interaction (Extended)	20	-
• A big design-focussed group project • Also had 2 class tests (multiple choice) and an essay based on the design project • Has a lot of focus on the design process for programs, but also covers the senses and how they need to be considered in design, testing and ethnography • Generally has a couple of readings each week • Weekly readings are generally written in a way that's scientific, but also very interesting and practical, making it easy and quick to read • Take this module if you care about the social issues around computer science, and want to learn how to make your programs more accessible to the general public • Also if you want to work on making commercial programs			

已通过 Google 文档发布 举报滥用行为 了解详情

- You get to pick the groups
  - How many people in a group?
    - About 5/6?
- Didn't really seem to have any proper content - the module consisted almost entirely of waffle like soft eng
  - If you don't like software engineering you probably won't like this
- It's honestly not too bad but if you're the kind of person who wants things to follow logically and sensibly this module with frustrate you a lot
- Warning that the ethnography portion is lowkey racist

06-30227	Intelligent Robotics	20	-
06-30244	Intelligent Robotics (Extended)	20	-

- Teamwork heavy - Great fun if you're with people you know but can be a faceburden (especially if virtual)
- Has a bad reputation as requiring you to do all nighters when it's not COVID, which isn't entirely undeserved
  - If you're willing to do one or two all-nighters, this can be a positive but obviously it isn't a given. You will need to sacrifice a lot of time to set up and debugging rather than just writing code
- If you enjoyed robotics in first year it's more of the same, but you're not using lego robots any more
  - If it's in person it'll be these big red circular robots you can stick your laptop on. They're kinda cute
- The Extended version is very maths heavy. Lecturer assumes you know everything you need (calculus, advanced statistics) and that you will put in the extra

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

06-30255	Machine Learning and Intelligent Data Analysis (Extended)	20	-
• Covers a lot of content already touched on in AI 1&2, just at a deeper level. Very theoretical and abstract. • It is very mathsy. • Complements other modules such as Neural or Robotics well • Numerous errors in slides and corrections makes things confusing • This year (2020/2021), we did weekly summative quizzes, which contributed 2.5% to the overall mark. This was a good way to solidify your knowledge.			
06-29289	Mobile & Ubiquitous Computing	-	20
06-25689	Mobile & Ubiquitous Computing (Extended)	-	20
• Sandy is great <3 ✨ ✨ • Before the pandemic and new teaching format involved an essay and programming a mobile app - This year Sandy put effort into trying to make lectures engaging/different during the pandemic, and so he recorded podcasts (!!!) for a lot of the required reading as well as module notes • ✨ Weekly Kahoot Quizzes ✨ (unironically, the best) • Lectures are about aspects of design relating to mobile and ubiquitous applications such as notifications, location services and input methods, as well as touching on some of the social issues re: Ubiquitous Computing (eg. security and privacy) • Labs are learning how to use Android Studio (note that this program is somewhat memory intensive) - no seriously, it's			

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

	each week that contain assessed content, usually the papers aren't too bad to parse though and he does provide summaries - You should be reading them for the best understanding. • Had the best feedback of any module we've seen on the student rep surveys (which it deserves, Sandy's a great lecturer) • SANDY MY BELOVED <3 • Probably the easiest module you will do on this course, **though this may change next year as Sandy is leaving.** Programming tasks are basic at their hardest. Written questions may be challenging if you aren't used to them.		
06-32167	Neural Computation	20	-
06-32212	Neural Computation (Extended)	20	-
	• This module is a difficult module • It's recommended to take it with Machine Learning as the crossover in topics will reduce work required • Involves a group project where you will write a python program that implements a neural network and write a report on it • In previous years, the coursework was way too difficult for the module and (despite being called out on it) the lecturer and school did nothing about it • Also has a huge difficulty spike about three weeks in		
06-30230	Programming Language Principles, Design, and Implementation	20	-
06-30256	Programming Language Principles, Design, and	20	-

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

rewarding, and you gain knowledge of how some *very* complex parts of CS work.
- Can be challenging but it is incredibly well organized which can make it more easily manageable than an easier module with a poorer structure.
- Each part of the module is very different and each lecturer has different styles.
    - Vincent's portion was very lambda-calculus oriented, notation-intense, and reasoned about formally defining language semantics and typing rules
    - Dan's portion focused on an overview of the pipeline that turns code into runnable programs - however, Dan's focus is mostly on functional programming languages, so it's good to be quite familiar with Haskell/others for this.
    - Dave's portion was about how to check program behaviour - super useful for understanding the tech behind static analysis (pretty common in most software engineering teams) and newer techniques such as symbolic execution.
- Expect to put quite a reasonable amount of time understanding the lectures beyond just the time listening to them - even though you're always recommended to do that, this is one module you definitely will need to.
- For assessments
    - It's helpful to be able to write LaTeX before it starts :D
    - The assessments can take a long *long* time to do and you will struggle

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

06-30257	Real-World Systems (Extended)	20	-

- If you enjoyed the work and exercises in your 2nd year security module, you'll hopefully really enjoy this <3
- Directly follows and extends the last thing you learn in 2nd Year Security (Buffer Overflows), so if you weren't comfortable with that, beware (or catch up first)
- Covers a lot of different areas, from software exploits, to protocol analysis, hardware side channel attacks
- If you don't have access to a linux machine, you are making it very difficult for yourself to do this module
    - Please at least dual-boot it's basically required
    - Some people used Windows Subsystem for Linux?
- Both Tom and David are *excellent* lecturers, and really up to date with their respective fields, so there's a few lectures on really experimental stuff, that wouldn't be talked about anywhere else
- Both lecturers and TA's answered/responded quickly, and were generally helpful :)
- Like the 2nd year module, the assessments often include problem-solving where you might have to do a lot of trial-and-error before getting anywhere. Also involves a lot of disassembly analysis (though that's mostly taught along the way)
- Also had a fun but challenging group project (you don't get to pick your teams) where you have to hack a game client and design your own cheats
- You DON'T have to be a security expert to understand the content. This was one of my fears before taking the module, but it was unjustified.

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

06-30208	Computer Science in Schools	-	20
06-30235	Teaching Computer Science in Schools (Extended)	-	20

• In normal times, going into local schools 1 day a week - remember normal teaching hours!
• Limited amount of students can take this
• Requires you do lectures in first term as well as the teaching in second term
• Student rep feedback: Very highly rated module but only a small handful of students take it.
  ○ Only 10 could take it this year due to COVID reducing placement numbers - you can be taken off if they don't have a placement for you
  ○ You'll know early in first term if you can't do it due to numbers - they'll ask you to do a quick assignment explaining why you want to do the module (I personally got taken off due to risk I wouldn't be able to do the placement due to shielding)			

Optional Modules (LM)			
06-34233	Forensics, Malware, and Penetration Testing	-	20
• Is a combination of the past modules "Forensics and Malware" and "Penetration Testing". • The same great lecturers as Security of Real-World Systems (see above) and covers a lot of the same content + a lot more, especially a lot of new stuff on			

已通过 Google 文档发布 举报滥用行为 了解详情

General Advice

- General advice about doing a 3:1 split against the School's recommendation:
    - It *is* possible and you can do it successfully - keep in mind 100% CW modules if you want to make exam season easier on yourself
    - Can be useful (and most likely better) if the majority of good modules offered are in the first term.
    - Very few people who have done it recommend it to other people (because it is hell)
    - Avoid doing it if:
        - You have other ongoing weekly commitments (for example, running a committee!) - unless you **can handle the stress and workload**
        - You are a final year MSci Student. You can't save project work to next term with the presentation at the end of first term
        - You get overwhelmed easily by a large course load
        - You're planning on

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

Robotics
    - Remember your supervisor will still expect you to do some project work each week - being busy due to unbalanced course load isn't an excuse (so really it's more like a 4:2 split)
        - However, you will be able to get away with doing less work in the term with fewer credits - as long as you continue to make some progress each week.
- Taking a masters module in fourth year:
    - Technically you can ask nicely to do other masters modules if you're a fourth year.
        - To do this, email the office and they'll point you towards the person in charge of deciding it (probably Ian Batten)
    - This will be more successful if you can explain why, and sometimes they just won't let you, but you can try!
        - Especially if you can link the content of the module with the area you're doing your project in
        - In addition it helps if you can justify it

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

module if you're hoping to go into research)
    - Even if you can't take the module, you may be able to ask the lecturer to be added to the canvas course as an observer, so you can follow along with lectures/slides/etc, without actually taking the module.
- Managing project schedules
    - To avoid unnecessary stress with a project, **try and complete something every week**.
        - A neat technique for this is to keep a todo list/kanban board - always have an "easy/fun" task on it that's different from most of the other work you have to do. If you don't feel like working on a specific day, then you can pick up the one you've been saving.
    - Your supervisor doesn't necessarily have to be an expert in the field you're doing your project in, but it helps **a lot**
        - Don't be afraid of doing a project outside their specialisation, but definitely *ask* first

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

to do you may be able to switch - do this sooner rather than later (We were given the choice to try change if possible if you were unhappy)

- As you write logs/attend supervisor meetings, try and have accomplished at least one new thing each week. It doesn't matter if it's just a small refactor, writing some documentation, or a really small feature, it's important to get into the habit of doing something.
- Also be honest with your supervisor. They're there to help you, not judge you.
- Falling behind on project can *really* hurt you later in the year, and making constant (even if slow) progress, is the best way to stop this from happening.
    - Try to keep ahead if anything, because then if you do have a week where you just can't get stuff done (because of illness or whatever), a) it reduces the stress for yourself and b) your supervisor will be a lot more understanding

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

vaguely-defined problem area, etc), talk to your supervisor asap!
        - They will be able to help you plan next-steps whatever those happen to be - either helping you develop your idea, finding you help to get you unstuck, or even working out a new idea that you're more invested in.
    - If possible, you want to have your project topic locked in for your proposal, however, you can still change it after that if you need to.
    - Despite the fact that this feels kinda shit to do, be aware that *lots* of people have and will change their project direction slightly or even completely throughout the year.
- Having weird supervisor situations
    - You won't end up with a supervisor outside of CS or multiple supervisors without actively trying for it
    - However you can end up with two supervisors or a supervisor from another school by getting up to shenanigans
        - After supervisors are allocated,

已通过 Google 文档发布 举报滥用行为 了解详情

university (in our year that was psychology and sports science)
        - You have to sort these ones out yourself by having meetings with them and then emailing Sandy about it
        - I don't know how you get two supervisors (despite having two supervisors) you just make the right noises and it just… happens
    - Remember that if you take a supervisor from another school, you'll be on your own with regards to school policies for final year projects - they won't know what the school does, they only know their school's policies
    - If you have two supervisors, it will involve juggling
        - They will say that you should all have a meeting together. This won't happen
    - Genuinely can be worth it though, if they're covering a topic that you're interested in
        - Oddly enough they're quite good for

已通过 Google 文档发布 举报滥用行为 了解详情

a while going (on Zoom) to psychology paper discussions because my supervisor invited me
    - You will be very confused about your official status in terms of supervisors. You have to be okay with bureaucratic chaos.
- Worth mentioning that with the Coursework Assessment Week changes that a lot of the talk about assessment may not be relevant
    - Who knows what's going on with this really.

Maths students advice

- First off: I'm sorry
- The comp sci department Can and Will forget you exist. Make sure you have at least a couple friends who do flat CS because sometimes you just won't be told things
- You need to get permission from BOTH departments if you're splitting the course unevenly, so honestly most of the time it's just not worth it
- Double check what modules you're ALLOWED to take. They have advertised it wrong before, they have changed it at the last second before, they are definitely willing to do it again
- You only get a tutor in maths. If you do the maths project, they will also be your supervisor, whether they know anything about the topic or not. If you do the CS

已通过 Google 文档发布 举报滥用行为 了解详情

Module Choices 2021

每 5 分钟自动更新一次

need it

- You are as MUCH a CS student as the flat CS folks. You are entitled to all the same privileges and benefits. Don’t let anyone tell you otherwise